Supplementary Material

Trustworthy prediction intervals after moving an NIR calibration to a new instrument or season

Every table and figure in this document was generated automatically from the result files of the study. Method names: TJ+N, transfer jackknife+ with T^2/Q scaling; TJ+, without scaling; TJ+B, leave-one-batch-out version; Split, half of the standards for transfer and half for calibration (SCP-T-split); Gauss-LOO, t-interval from leave-one-out errors.

Table S1. Prediction error after transfer

Table S1. Mean RMSEP of the transferred predictor for each pipeline and number of standards m (corn averaged over mp5 and mp6). NONE: original calibration applied without transfer.

Data set	Response	DS (m=10)	DS (m=20)	NONE (m=10)	NONE (m=20)	PDS (m=10)	PDS (m=20)	SBC (m=10)	SBC (m=20)
Corn	moisture	0.308	0.238	1.598	1.592	0.349	0.304	0.251	0.229
Corn	oil	0.187	0.145	0.287	0.286	0.142	0.109	0.091	0.084
Corn	protein	0.439	0.316	0.794	0.794	0.233	0.194	0.168	0.158
Corn	starch	0.885	0.669	1.901	1.897	0.568	0.466	0.406	0.377
Mango	dry matter	–	–	1.052	1.052	–	–	1.155	1.095
Tablets	assay	9.216	5.998	7.351	7.347	6.434	5.101	7.797	7.298
Tablets	hardness	1.011	1.014	1.055	1.054	1.021	1.022	1.074	0.998
Tablets	weight	4.138	4.165	4.456	4.456	4.194	4.201	4.212	3.926

Table S2. Three confidence levels

Table S2. Number of scenarios flagged as too narrow / scenarios, with the mean percentage of repetitions that returned unbounded intervals in parentheses (main pipelines, random standards).

Method	$\alpha = 0.05$ (nominal 0.95)	$\alpha = 0.1$ (nominal 0.90)	$\alpha = 0.2$ (nominal 0.80)
TJ+N	0/40 (50%)	0/40 (0%)	0/40 (0%)
TJ+	0/40 (50%)	0/40 (0%)	0/40 (0%)
SCP-T-split-N	0/40 (90%)	0/40 (50%)	0/40 (0%)
Gauss-LOO	14/40 (0%)	0/40 (0%)	2/40 (0%)
ASTM-PI	29/40 (0%)	29/40 (0%)	29/40 (0%)
SCP-S	29/40 (0%)	33/40 (0%)	34/40 (0%)
WCP-S	29/40 (69%)	33/40 (49%)	33/40 (30%)
LF-conv	25/36 (0%)	20/36 (0%)	13/36 (0%)

Table S3. Components of TJ+N

Table S3. Ablation of TJ+N (α = 0.10, main pipelines). TCV+N: five-fold CV+; TJ-naive-N: full-fit centre with leave-one-out quantile; TJ-insample-N: residuals of the transfer fitted on all standards; TJ+N[T2], TJ+N[Q]: scaling with one diagnostic. Rel. IS: median interval score relative to TJ+N. Last column: coverage in the lowest/highest third of σ(x).

Variant	Corn cov.	Corn under	Corn rel. IS	Tablets cov.	Tablets under	Tablets rel. IS	Mango cov.	Mango under	Mango rel. IS	Cov. low/high σ tertile
TJ+N	0.931	0/24	1.000	0.910	0/12	1.000	0.904	0/4	1.000	0.904 / 0.933
TJ+	0.933	0/24	1.019	0.911	0/12	1.062	0.904	0/4	1.016	0.957 / 0.878
TCV+N	0.942	0/24	1.066	0.917	0/12	1.000	0.908	0/4	1.009	0.916 / 0.941
TJ-naive-N	0.919	0/24	0.961	0.905	0/12	0.999	0.903	0/4	0.996	0.894 / 0.924
TJ-insample-N	0.897	10/24	0.935	0.912	0/12	1.000	0.863	4/4	0.992	0.878 / 0.909
TJ+N[T2]	0.931	0/24	0.976	0.911	0/12	0.999	0.904	0/4	1.005	0.922 / 0.916
TJ+N[Q]	0.931	0/24	1.020	0.911	0/12	1.011	0.904	0/4	1.006	0.914 / 0.924

Table S4. Pipelines, directions and model types

Table S4. Reverse direction, ridge regression and a 1D-CNN (α = 0.10). Each cell: mean coverage (scenarios too narrow / scenarios); median interval score.

Setting	TJ+N	TJ+	Gauss-LOO	SCP-T-split-N	ASTM-PI	SCP-S	WCP-S	MC-dropout
Corn mp5→m5, PLS, PDS	0.931 (0/12); 1.11	0.932 (0/12); 1.21	0.905 (0/12); 1.17	0.971 (0/12); inf	0.657 (12/12); 1.34	0.757 (12/12); 1.25	0.800 (12/12); 1.53	–
Corn m5→mp5, ridge, PDS	0.943 (0/4); 1.21	0.953 (0/4); 1.17	0.904 (0/4); 1.06	1.000 (0/4); inf	0.057 (4/4); 3.6	0.416 (4/4); 1.98	0.518 (4/4); inf	–
Tablets assay, ridge, PDS	0.916 (0/1); 21.8	0.915 (0/1); 21.9	0.914 (0/1); 22	0.918 (0/1); 26.4	0.905 (0/1); 20.7	0.874 (1/1); 20.9	0.888 (1/1); 21.8	–
Mango, ridge, SBC	0.898 (0/1); 4.5	0.899 (0/1); 4.57	0.899 (0/1); 4.52	0.902 (0/1); 4.97	0.729 (1/1); 5.1	0.716 (1/1); 5.2	0.742 (1/1); inf	–
Mango, 1D-CNN, SBC	0.910 (0/4); 5.24	0.909 (0/4); 5.51	0.894 (2/4); 5.42	0.945 (0/4); 5.59	0.714 (4/4); 6.48	0.693 (4/4); 6.71	0.780 (4/4); inf	0.807 (4/4); 5.66

Table S5. Robustness

Table S5. Robustness (α = 0.10; corn m = 15, tablets assay and mango m = 20; 200 repetitions): mean and minimum coverage over scenarios, median interval score and scenarios flagged as too narrow.

dataset	variant	method	coverage	coverage_min	IS_median	n_under	n_scen
corn	half_source	ASTM-PI	0.335	0.071	2.841	4	4
corn	half_source	Gauss-LOO	0.908	0.898	1.092	0	4
corn	half_source	SCP-S	0.608	0.157	1.627	4	4
corn	half_source	SCP-T-split-N	1.0	1.0	inf	0	4
corn	half_source	TJ+	0.947	0.941	1.183	0	4
corn	half_source	TJ+N	0.948	0.932	1.2	0	4
corn	half_source	WCP-S	0.7	0.355	inf	4	4
corn	ks	ASTM-PI	0.232	0.033	2.929	4	4
corn	ks	Gauss-LOO	0.913	0.9	1.048	0	4
corn	ks	SCP-S	0.452	0.067	1.813	4	4
corn	ks	SCP-T-split-N	1.0	1.0	inf	0	4
corn	ks	TJ+	0.955	0.946	1.191	0	4
corn	ks	TJ+N	0.947	0.934	1.185	0	4
corn	ks	WCP-S	0.554	0.221	inf	4	4
corn	noise_100	ASTM-PI	0.181	0.033	3.558	4	4
corn	noise_100	Gauss-LOO	0.908	0.893	1.167	0	4
corn	noise_100	SCP-S	0.327	0.058	3.197	4	4
corn	noise_100	SCP-T-split-N	1.0	1.0	inf	0	4
corn	noise_100	TJ+	0.956	0.946	1.306	0	4
corn	noise_100	TJ+N	0.956	0.939	1.261	0	4
corn	noise_100	WCP-S	0.473	0.238	inf	4	4
corn	noise_1000	ASTM-PI	0.234	0.037	2.983	4	4
corn	noise_1000	Gauss-LOO	0.905	0.892	1.044	0	4
corn	noise_1000	SCP-S	0.456	0.066	1.794	4	4
corn	noise_1000	SCP-T-split-N	1.0	1.0	inf	0	4
corn	noise_1000	TJ+	0.951	0.943	1.17	0	4
corn	noise_1000	TJ+N	0.946	0.934	1.175	0	4
corn	noise_1000	WCP-S	0.554	0.221	inf	4	4
corn	noise_300	ASTM-PI	0.23	0.036	3.006	4	4
corn	noise_300	Gauss-LOO	0.906	0.895	1.054	0	4
corn	noise_300	SCP-S	0.446	0.065	1.97	4	4
corn	noise_300	SCP-T-split-N	1.0	1.0	inf	0	4
corn	noise_300	TJ+	0.952	0.944	1.175	0	4
corn	noise_300	TJ+N	0.945	0.931	1.168	0	4
corn	noise_300	WCP-S	0.548	0.22	inf	4	4
corn	shift_-1	ASTM-PI	0.086	0.0	18.75	4	4
corn	shift_-1	Gauss-LOO	0.17	0.025	13.629	4	4
corn	shift_-1	SCP-S	0.074	0.0	19.4	4	4
corn	shift_-1	SCP-T-split-N	1.0	1.0	inf	0	4
corn	shift_-1	TJ+	0.286	0.058	9.399	4	4
corn	shift_-1	TJ+N	0.843	0.679	5.656	2	4
corn	shift_-1	WCP-S	0.416	0.361	30.404	4	4
corn	shift_1	ASTM-PI	0.079	0.0	19.718	4	4

3

dataset	variant	method	coverage	coverage_min	IS_median	n_under	n_scen
corn	shift_1	Gauss-LOO	0.151	0.02	14.643	4	4
corn	shift_1	SCP-S	0.067	0.0	20.582	4	4
corn	shift_1	SCP-T-split-N	1.0	1.0	inf	0	4
corn	shift_1	TJ+	0.262	0.05	10.341	4	4
corn	shift_1	TJ+N	0.823	0.605	5.795	2	4
corn	shift_1	WCP-S	0.497	0.416	inf	4	4
mango	half_source	ASTM-PI	0.755	0.755	4.864	1	1
mango	half_source	Gauss-LOO	0.899	0.899	4.46	0	1
mango	half_source	SCP-S	0.752	0.752	4.887	1	1
mango	half_source	SCP-T-split-N	0.905	0.905	4.857	0	1
mango	half_source	TJ+	0.9	0.9	4.51	0	1
mango	half_source	TJ+N	0.899	0.899	4.479	0	1
mango	half_source	WCP-S	0.785	0.785	inf	1	1
mango	ks	ASTM-PI	0.752	0.752	4.888	1	1
mango	ks	Gauss-LOO	0.968	0.968	4.994	0	1
mango	ks	SCP-S	0.733	0.733	5.028	1	1
mango	ks	SCP-T-split-N	0.956	0.956	5.337	0	1
mango	ks	TJ+	0.968	0.968	5.005	0	1
mango	ks	TJ+N	0.964	0.964	4.958	0	1
mango	ks	WCP-S	0.739	0.739	inf	1	1
mango	noise_100	ASTM-PI	0.508	0.508	13.075	1	1
mango	noise_100	Gauss-LOO	0.895	0.895	7.863	0	1
mango	noise_100	SCP-S	0.49	0.49	13.545	1	1
mango	noise_100	SCP-T-split-N	0.91	0.91	8.261	0	1
mango	noise_100	TJ+	0.9	0.9	7.968	0	1
mango	noise_100	TJ+N	0.901	0.901	7.627	0	1
mango	noise_100	WCP-S	0.507	0.507	inf	1	1
mango	noise_1000	ASTM-PI	0.744	0.744	5.038	1	1
mango	noise_1000	Gauss-LOO	0.9	0.9	4.554	0	1
mango	noise_1000	SCP-S	0.726	0.726	5.186	1	1
mango	noise_1000	SCP-T-split-N	0.908	0.908	4.957	0	1
mango	noise_1000	TJ+	0.902	0.902	4.63	0	1
mango	noise_1000	TJ+N	0.902	0.902	4.54	0	1
mango	noise_1000	WCP-S	0.733	0.733	inf	1	1
mango	noise_300	ASTM-PI	0.693	0.693	6.271	1	1
mango	noise_300	Gauss-LOO	0.9	0.9	5.235	0	1
mango	noise_300	SCP-S	0.675	0.675	6.478	1	1
mango	noise_300	SCP-T-split-N	0.91	0.91	5.722	0	1
mango	noise_300	TJ+	0.907	0.907	5.328	0	1
mango	noise_300	TJ+N	0.908	0.908	5.149	0	1
mango	noise_300	WCP-S	0.685	0.685	inf	1	1
mango	shift_-1	ASTM-PI	0.023	0.023	50.584	1	1
mango	shift_-1	Gauss-LOO	0.07	0.07	42.187	1	1
mango	shift_-1	SCP-S	0.018	0.018	52.878	1	1
mango	shift_-1	SCP-T-split-N	0.789	0.789	13.758	1	1
mango	shift_-1	TJ+	0.081	0.081	41.036	1	1
mango	shift_-1	TJ+N	0.732	0.732	13.493	1	1

4

dataset	variant	method	coverage	coverage_min	IS_median	n_under	n_scen
mango	shift_-1	WCP-S	0.305	0.305	inf	1	1
mango	shift_1	ASTM-PI	0.109	0.109	30.41	1	1
mango	shift_1	Gauss-LOO	0.206	0.206	25.282	1	1
mango	shift_1	SCP-S	0.069	0.069	34.439	1	1
mango	shift_1	SCP-T-split-N	0.97	0.97	15.736	0	1
mango	shift_1	TJ+	0.216	0.216	24.705	1	1
mango	shift_1	TJ+N	0.974	0.974	14.14	0	1
mango	shift_1	WCP-S	0.207	0.207	inf	1	1
tablets	half_source	ASTM-PI	0.971	0.971	21.998	0	1
tablets	half_source	Gauss-LOO	0.916	0.916	20.21	0	1
tablets	half_source	SCP-S	0.957	0.957	20.65	0	1
tablets	half_source	SCP-T-split-N	0.913	0.913	22.635	0	1
tablets	half_source	TJ+	0.908	0.908	20.01	0	1
tablets	half_source	TJ+N	0.907	0.907	19.729	0	1
tablets	half_source	WCP-S	0.964	0.964	21.402	0	1
tablets	ks	ASTM-PI	0.94	0.94	21.237	0	1
tablets	ks	Gauss-LOO	0.954	0.954	23.123	0	1
tablets	ks	SCP-S	0.922	0.922	21.252	0	1
tablets	ks	SCP-T-split-N	0.92	0.92	25.461	0	1
tablets	ks	TJ+	0.956	0.956	23.808	0	1
tablets	ks	TJ+N	0.93	0.93	20.831	0	1
tablets	ks	WCP-S	0.928	0.928	21.95	0	1
tablets	noise_100	ASTM-PI	0.927	0.927	21.563	0	1
tablets	noise_100	Gauss-LOO	0.91	0.91	23.167	0	1
tablets	noise_100	SCP-S	0.91	0.91	21.729	0	1
tablets	noise_100	SCP-T-split-N	0.913	0.913	28.094	0	1
tablets	noise_100	TJ+	0.913	0.913	23.109	0	1
tablets	noise_100	TJ+N	0.914	0.914	22.03	0	1
tablets	noise_100	WCP-S	0.926	0.926	23.533	0	1
tablets	noise_1000	ASTM-PI	0.928	0.928	21.591	0	1
tablets	noise_1000	Gauss-LOO	0.91	0.91	23.064	0	1
tablets	noise_1000	SCP-S	0.91	0.91	21.695	0	1
tablets	noise_1000	SCP-T-split-N	0.916	0.916	27.687	0	1
tablets	noise_1000	TJ+	0.913	0.913	23.01	0	1
tablets	noise_1000	TJ+N	0.916	0.916	22.036	0	1
tablets	noise_1000	WCP-S	0.927	0.927	23.478	0	1
tablets	noise_300	ASTM-PI	0.927	0.927	21.568	0	1
tablets	noise_300	Gauss-LOO	0.91	0.91	23.077	0	1
tablets	noise_300	SCP-S	0.91	0.91	21.718	0	1
tablets	noise_300	SCP-T-split-N	0.916	0.916	27.932	0	1
tablets	noise_300	TJ+	0.912	0.912	23.032	0	1
tablets	noise_300	TJ+N	0.915	0.915	22.011	0	1
tablets	noise_300	WCP-S	0.926	0.926	23.397	0	1
tablets	shift_-1	ASTM-PI	0.02	0.02	304.33	1	1
tablets	shift_-1	Gauss-LOO	0.035	0.035	300.236	1	1
tablets	shift_-1	SCP-S	0.015	0.015	314.736	1	1
tablets	shift_-1	SCP-T-split-N	0.253	0.253	226.108	1	1

5

dataset	variant	method	coverage	coverage_min	IS_median	n_under	n_scen
tablets	shift_-1	TJ+	0.034	0.034	286.928	1	1
tablets	shift_-1	TJ+N	0.069	0.069	259.707	1	1
tablets	shift_-1	WCP-S	0.022	0.022	340.682	1	1
tablets	shift_1	ASTM-PI	0.008	0.008	350.126	1	1
tablets	shift_1	Gauss-LOO	0.018	0.018	347.84	1	1
tablets	shift_1	SCP-S	0.007	0.007	360.886	1	1
tablets	shift_1	SCP-T-split-N	0.22	0.22	245.495	1	1
tablets	shift_1	TJ+	0.014	0.014	341.348	1	1
tablets	shift_1	TJ+N	0.04	0.04	299.11	1	1
tablets	shift_1	WCP-S	0.015	0.015	444.978	1	1

Table S6. Transfer between handheld fruit spectrometers

Table S6. Mango instrument transfer, source unit 15006 to three target units, four values of m (12 scenarios per pipeline; α = 0.10; 200 repetitions; standards drawn at random).

Pipeline	Method	Coverage (range)	Under	Unbounded (% reps)	Median width (% DM)	RMSEP (% DM)
PDS	TJ+N	0.931 (0.91–0.97)	0/12	0	11.05	4.03
PDS	TJ+	0.929 (0.90–0.97)	0/12	0	12.79	4.03
PDS	SCP-T-split-N	0.943 (0.91–1.00)	0/12	25	11.15	4.03
PDS	Gauss-LOO	0.916 (0.90–0.93)	0/12	0	12.13	4.03
PDS	ASTM-PI	0.355 (0.22–0.51)	12/12	0	2.71	4.03
PDS	SCP-S	0.342 (0.20–0.50)	12/12	0	2.60	4.03
PDS	WCP-S	0.292 (0.15–0.47)	12/12	25	2.24	4.03
PDS	Transport	0.978 (0.95–0.99)	0/12	0	24.63	4.03
PDS	LF-conv	0.887 (0.86–0.91)	6/12	0	11.83	4.03
SBC	TJ+N	0.906 (0.90–0.92)	0/12	0	4.00	1.14
SBC	TJ+	0.906 (0.90–0.92)	0/12	0	3.87	1.14
SBC	SCP-T-split-N	0.941 (0.90–1.00)	0/12	25	4.56	1.14
SBC	Gauss-LOO	0.900 (0.89–0.91)	0/12	0	3.85	1.14
SBC	ASTM-PI	0.781 (0.74–0.81)	12/12	0	2.66	1.14
SBC	SCP-S	0.762 (0.73–0.80)	12/12	0	2.60	1.14
SBC	WCP-S	1.000 (1.00–1.00)	0/12	100	4.54	1.14

6

Table S7. Gaussian intervals with difficulty scaling

Table S7. TJ+N compared with Gaussian leave-one-out intervals without (Gauss-LOO) and with (Gauss-LOO-N) σ(x) scaling (200 repetitions).

Data/pipeline set	Method	α = 0.05	α = 0.10	α = 0.20
corn	TJ+N	0.986 (0/24)	0.932 (0/24)	0.835 (0/24)
corn	Gauss-LOO	0.945 (6/24)	0.903 (0/24)	0.818 (0/24)
corn	Gauss-LOO-N	0.951 (0/24)	0.905 (0/24)	0.811 (0/24)
mango	TJ+N	0.968 (0/4)	0.902 (0/4)	0.803 (0/4)
mango	Gauss-LOO	0.955 (0/4)	0.902 (0/4)	0.792 (1/4)
mango	Gauss-LOO-N	0.956 (0/4)	0.902 (0/4)	0.790 (2/4)
mango_inst	TJ+N	0.974 (0/24)	0.919 (0/24)	0.827 (0/24)
mango_inst	Gauss-LOO	0.950 (4/24)	0.908 (0/24)	0.820 (0/24)
mango_inst	Gauss-LOO-N	0.958 (6/24)	0.915 (0/24)	0.810 (3/24)
tablets	TJ+N	0.971 (0/12)	0.912 (0/12)	0.820 (0/12)
tablets	Gauss-LOO	0.949 (6/12)	0.910 (0/12)	0.825 (1/12)
tablets	Gauss-LOO-N	0.948 (2/12)	0.907 (1/12)	0.819 (0/12)

Table S8. Quarter-channel drift

Table S8. Coverage under a ±0.25-channel wavelength shift of the routine spectra after standardisation (α = 0.10; 200 repetitions).

dataset	shift	method	coverage	coverage_min	too_narrow	n
corn	shift_-0.25	ASTM-PI	0.152	0.017	4	4
corn	shift_-0.25	Gauss-LOO	0.692	0.672	4	4
corn	shift_-0.25	SCP-S	0.259	0.025	4	4
corn	shift_-0.25	SCP-T-split-N	1.0	1.0	0	4
corn	shift_-0.25	TJ+	0.807	0.781	4	4
corn	shift_-0.25	TJ+N	0.928	0.879	0	4
corn	shift_-0.25	WCP-S	0.459	0.281	4	4
corn	shift_0.25	ASTM-PI	0.152	0.021	4	4
corn	shift_0.25	Gauss-LOO	0.665	0.616	4	4
corn	shift_0.25	SCP-S	0.251	0.03	4	4
corn	shift_0.25	SCP-T-split-N	1.0	1.0	0	4
corn	shift_0.25	TJ+	0.781	0.746	4	4
corn	shift_0.25	TJ+N	0.919	0.839	1	4
corn	shift_0.25	WCP-S	0.501	0.392	4	4
mango	shift_-0.25	ASTM-PI	0.624	0.624	1	1
mango	shift_-0.25	Gauss-LOO	0.781	0.781	1	1
mango	shift_-0.25	SCP-S	0.605	0.605	1	1
mango	shift_-0.25	SCP-T-split-N	0.885	0.885	0	1
mango	shift_-0.25	TJ+	0.786	0.786	1	1
mango	shift_-0.25	TJ+N	0.861	0.861	1	1
mango	shift_-0.25	WCP-S	0.619	0.619	1	1
mango	shift_0.25	ASTM-PI	0.682	0.682	1	1
mango	shift_0.25	Gauss-LOO	0.833	0.833	1	1
mango	shift_0.25	SCP-S	0.659	0.659	1	1
mango	shift_0.25	SCP-T-split-N	0.958	0.958	0	1
mango	shift_0.25	TJ+	0.832	0.832	1	1
mango	shift_0.25	TJ+N	0.961	0.961	0	1

7

dataset	shift	method	coverage	coverage_min	too_narrow	n
mango	shift_0.25	WCP-S	0.682	0.682	1	1
tablets	shift_-0.25	ASTM-PI	0.663	0.663	1	1
tablets	shift_-0.25	Gauss-LOO	0.659	0.659	1	1
tablets	shift_-0.25	SCP-S	0.613	0.613	1	1
tablets	shift_-0.25	SCP-T-split-N	0.697	0.697	1	1
tablets	shift_-0.25	TJ+	0.668	0.668	1	1
tablets	shift_-0.25	TJ+N	0.591	0.591	1	1
tablets	shift_-0.25	WCP-S	0.67	0.67	1	1
tablets	shift_0.25	ASTM-PI	0.661	0.661	1	1
tablets	shift_0.25	Gauss-LOO	0.658	0.658	1	1
tablets	shift_0.25	SCP-S	0.613	0.613	1	1
tablets	shift_0.25	SCP-T-split-N	0.726	0.726	1	1
tablets	shift_0.25	TJ+	0.675	0.675	1	1
tablets	shift_0.25	TJ+N	0.615	0.615	1	1
tablets	shift_0.25	WCP-S	0.671	0.671	1	1

Table S9. Transfer and model settings

Table S9. Sensitivity to the PDS window (hw1 = 3 channels, hw3 = 7, hw5 = 11; main analysis 5) and to the number of PLS components (±2).

dataset	setting	method	coverage	coverage_min	IS_median	rmsep	n_under	n_scen
corn	hw1	ASTM-PI	0.25	0.033	2.655	0.237	4	4
corn	hw1	Gauss-LOO	0.903	0.89	0.932	0.237	0	4
corn	hw1	SCP-S	0.496	0.064	1.566	0.237	4	4
corn	hw1	SCP-T-split-N	1.0	1.0	inf	0.237	0	4
corn	hw1	TJ+	0.941	0.93	1.057	0.237	0	4
corn	hw1	TJ+N	0.941	0.927	1.083	0.237	0	4
corn	hw1	WCP-S	0.588	0.217	inf	0.237	4	4
corn	hw3	ASTM-PI	0.213	0.028	3.314	0.348	4	4
corn	hw3	Gauss-LOO	0.909	0.896	1.21	0.348	0	4
corn	hw3	SCP-S	0.404	0.056	2.363	0.348	4	4
corn	hw3	SCP-T-split-N	1.0	1.0	inf	0.348	0	4
corn	hw3	TJ+	0.962	0.951	1.313	0.348	0	4
corn	hw3	TJ+N	0.954	0.938	1.272	0.348	0	4
corn	hw3	WCP-S	0.509	0.217	inf	0.348	4	4
corn	hw5	ASTM-PI	0.17	0.032	4.362	0.503	4	4
corn	hw5	Gauss-LOO	0.903	0.896	1.549	0.503	0	4
corn	hw5	SCP-S	0.314	0.053	4.002	0.503	4	4
corn	hw5	SCP-T-split-N	1.0	1.0	inf	0.503	0	4
corn	hw5	TJ+	0.967	0.955	1.735	0.503	0	4
corn	hw5	TJ+N	0.962	0.95	1.512	0.503	0	4
corn	hw5	WCP-S	0.44	0.233	inf	0.503	4	4
corn	lv+2	ASTM-PI	0.189	0.033	3.136	0.283	4	4
corn	lv+2	Gauss-LOO	0.905	0.893	1.047	0.283	0	4
corn	lv+2	SCP-S	0.436	0.059	1.872	0.283	4	4
corn	lv+2	SCP-T-split-N	1.0	1.0	inf	0.283	0	4

dataset	setting	method	coverage	coverage_min	IS_median	rmsep	n_under	n_scen
corn	lv+2	TJ+	0.952	0.943	1.18	0.283	0	4
corn	lv+2	TJ+N	0.949	0.939	1.2	0.283	0	4
corn	lv+2	WCP-S	0.552	0.237	inf	0.283	4	4
corn	lv-2	ASTM-PI	0.313	0.051	2.488	0.275	4	4
corn	lv-2	Gauss-LOO	0.904	0.894	1.06	0.275	0	4
corn	lv-2	SCP-S	0.516	0.083	1.616	0.275	4	4
corn	lv-2	SCP-T-split-N	1.0	1.0	inf	0.275	0	4
corn	lv-2	TJ+	0.951	0.943	1.161	0.275	0	4
corn	lv-2	TJ+N	0.944	0.933	1.191	0.275	0	4
corn	lv-2	WCP-S	0.589	0.203	2.741	0.275	4	4
mango	lv+2	ASTM-PI	0.737	0.737	5.054	1.108	1	1
mango	lv+2	Gauss-LOO	0.898	0.898	4.538	1.108	0	1
mango	lv+2	SCP-S	0.721	0.721	5.186	1.108	1	1
mango	lv+2	SCP-T-split-N	0.902	0.902	4.989	1.108	0	1
mango	lv+2	TJ+	0.899	0.899	4.612	1.108	0	1
mango	lv+2	TJ+N	0.898	0.898	4.535	1.108	0	1
mango	lv+2	WCP-S	0.734	0.734	inf	1.108	1	1
mango	lv-2	ASTM-PI	0.757	0.757	5.016	1.106	1	1
mango	lv-2	Gauss-LOO	0.896	0.896	4.587	1.106	0	1
mango	lv-2	SCP-S	0.745	0.745	5.117	1.106	1	1
mango	lv-2	SCP-T-split-N	0.906	0.906	4.968	1.106	0	1
mango	lv-2	TJ+	0.9	0.9	4.642	1.106	0	1
mango	lv-2	TJ+N	0.899	0.899	4.544	1.106	0	1
mango	lv-2	WCP-S	0.793	0.793	inf	1.106	1	1
tablets	hw1	ASTM-PI	0.943	0.943	20.805	4.767	0	1
tablets	hw1	Gauss-LOO	0.911	0.911	21.754	4.767	0	1
tablets	hw1	SCP-S	0.928	0.928	20.589	4.767	0	1
tablets	hw1	SCP-T-split-N	0.912	0.912	23.102	4.767	0	1
tablets	hw1	TJ+	0.913	0.913	21.903	4.767	0	1
tablets	hw1	TJ+N	0.916	0.916	20.449	4.767	0	1
tablets	hw1	WCP-S	0.939	0.939	22.007	4.767	0	1
tablets	hw3	ASTM-PI	0.906	0.906	22.434	5.39	0	1
tablets	hw3	Gauss-LOO	0.91	0.91	24.192	5.39	0	1
tablets	hw3	SCP-S	0.884	0.884	22.73	5.39	1	1
tablets	hw3	SCP-T-split-N	0.92	0.92	33.539	5.39	0	1
tablets	hw3	TJ+	0.916	0.916	24.22	5.39	0	1
tablets	hw3	TJ+N	0.916	0.916	23.243	5.39	0	1
tablets	hw3	WCP-S	0.909	0.909	24.208	5.39	0	1
tablets	hw5	ASTM-PI	0.867	0.867	25.273	5.98	1	1
tablets	hw5	Gauss-LOO	0.912	0.912	27.218	5.98	0	1
tablets	hw5	SCP-S	0.837	0.837	26.267	5.98	1	1
tablets	hw5	SCP-T-split-N	0.915	0.915	37.461	5.98	0	1
tablets	hw5	TJ+	0.925	0.925	27.284	5.98	0	1
tablets	hw5	TJ+N	0.924	0.924	26.505	5.98	0	1
tablets	hw5	WCP-S	0.87	0.87	29.995	5.98	1	1
tablets	lv+2	ASTM-PI	0.898	0.898	22.363	5.326	0	1
tablets	lv+2	Gauss-LOO	0.908	0.908	24.135	5.326	0	1
tablets	lv+2	SCP-S	0.876	0.876	23.175	5.326	1	1

9

dataset	setting	method	coverage	coverage_min	IS_median	rmsep	n_under	n_scen
tablets	lv+2	SCP-T-split-N	0.918	0.918	26.722	5.326	0	1
tablets	lv+2	TJ+	0.91	0.91	24.74	5.326	0	1
tablets	lv+2	TJ+N	0.918	0.918	21.987	5.326	0	1
tablets	lv+2	WCP-S	0.907	0.907	inf	5.326	0	1
tablets	lv-2	ASTM-PI	0.903	0.903	29.591	6.999	0	1
tablets	lv-2	Gauss-LOO	0.909	0.909	31.299	6.999	0	1
tablets	lv-2	SCP-S	0.912	0.912	29.915	6.999	0	1
tablets	lv-2	SCP-T-split-N	0.908	0.908	37.834	6.999	0	1
tablets	lv-2	TJ+	0.912	0.912	32.295	6.999	0	1
tablets	lv-2	TJ+N	0.916	0.916	28.684	6.999	0	1
tablets	lv-2	WCP-S	0.92	0.92	30.799	6.999	0	1

Table S10. Computing time

Table S10. Wall-clock time and peak memory (one CPU core, median of 20 runs).

dataset	kind	m	n_test	n_channels	step	median_s	p90_s	peak_mem_MB	source_model_fit_s
corn	pds	20	10	700	point prediction only (1 transfer fit)	0.0046	0.0061	0.8785	0.1027
corn	pds	20	10	700	SCP-T-split (1 fit)	0.0047	0.0065	0.7389	0.1027
corn	pds	20	10	700	TJ+N incl. m LOO + 5 CV fits	0.1469	0.1572	1.2104	0.1027
tablets	pds	50	410	650	point prediction only (1 transfer fit)	0.0243	0.0298	14.3825	0.2743
tablets	pds	50	410	650	SCP-T-split (1 fit)	0.0259	0.0265	14.3831	0.2743
tablets	pds	50	410	650	TJ+N incl. m LOO + 5 CV fits	2.7857	2.9018	15.2063	0.2743
mango	sbc	50	675	103	point prediction only (1 transfer fit)	0.0004	0.0005	1.1246	1.4279
mango	sbc	50	675	103	SCP-T-split (1 fit)	0.0004	0.0004	1.1246	1.4279
mango	sbc	50	675	103	TJ+N incl. m LOO + 5 CV fits	0.0635	0.0742	3.1352	1.4279

Table S11. Batch designs, every combination

Table S11. Standards from K consignments × n_g fruit (α = 0.10; 300 repetitions). Coverage for fruit from consignments without standards (consignment-weighted); * flagged as too narrow (upper 95% bootstrap limit < 0.90); unb.: unbounded in more than half of the repetitions. Last two columns: TJ+N coverage for the remaining fruit of the sampled consignments, and the intra-consignment correlation (ICC) of the errors of the transferred predictor. Confidence intervals for every cell are in results/summary/E10_summary.csv.

Setting	Pipeline	Design	TJ+N	TJ+B	Gauss-LOO	Split	ANOVA-PI	Gauss-LGO	ASTM-PI	TJ+N, same batches	ICC
inst_15014	PDS	chrono 5×4	0.927	unb.	0.804*	0.903	0.870*	0.905	0.424*	0.933	0.62
inst_15014	PDS	2×10	0.897	unb.	0.733*	0.872*	0.891	0.984	0.335*	0.926	0.59
inst_15014	PDS	5×4	0.916	unb.	0.846*	0.894	0.916	0.945	0.364*	0.926	0.58
inst_15014	PDS	10×2	0.925	0.981	0.889*	0.901	0.925	0.926	0.363*	0.932	0.56
inst_15014	PDS	15×1	0.976	0.976	0.928	unb.	0.936	0.928	0.351*	0.977	0.55
inst_15014	PDS	20×1	0.943	0.943	0.921	0.914	0.928	0.921	0.382*	0.943	0.54
inst_15014	SBC	chrono 5×4	0.763*	unb.	0.755*	0.800*	0.848*	0.868*	0.757*	0.919	0.69
inst_15014	SBC	2×10	0.672*	unb.	0.662*	0.700*	0.910	0.997	0.643*	0.907	0.63
inst_15014	SBC	5×4	0.824*	unb.	0.822*	0.845*	0.925	0.927	0.719*	0.916	0.61
inst_15014	SBC	10×2	0.882*	0.947	0.876*	0.889	0.922	0.910	0.744*	0.921	0.60
inst_15014	SBC	15×1	0.940	0.940	0.903	unb.	0.920	0.903	0.748*	0.959	0.60
inst_15014	SBC	20×1	0.906	0.906	0.899	0.910	0.912	0.899	0.758*	0.926	0.59
inst_15016	PDS	chrono 5×4	0.932	unb.	0.775*	0.911	0.859*	0.889	0.394*	0.938	0.57
inst_15016	PDS	2×10	0.907	unb.	0.762*	0.875*	0.854*	0.971	0.291*	0.927	0.56
inst_15016	PDS	5×4	0.918	unb.	0.855*	0.904	0.923	0.951	0.254*	0.932	0.55
inst_15016	PDS	10×2	0.935	0.982	0.901	0.904	0.935	0.933	0.262*	0.940	0.55
inst_15016	PDS	15×1	0.968	0.968	0.920	unb.	0.929	0.920	0.263*	0.972	0.54
inst_15016	PDS	20×1	0.930	0.930	0.910	0.908	0.919	0.910	0.272*	0.933	0.55
inst_15016	SBC	chrono 5×4	0.919	unb.	0.877*	0.923	0.923	0.943	0.807*	0.908	0.43
inst_15016	SBC	2×10	0.774*	unb.	0.753*	0.799*	0.934	0.999	0.705*	0.906	0.47
inst_15016	SBC	5×4	0.871*	unb.	0.859*	0.874*	0.938	0.945	0.760*	0.915	0.46
inst_15016	SBC	10×2	0.891	0.950	0.886*	0.900	0.922	0.913	0.770*	0.917	0.46
inst_15016	SBC	15×1	0.936	0.936	0.898	unb.	0.916	0.898	0.771*	0.948	0.46
inst_15016	SBC	20×1	0.900	0.900	0.899	0.906	0.913	0.899	0.783*	0.911	0.46
inst_15029	PDS	chrono 5×4	0.937	unb.	0.799*	0.909	0.792*	0.908	0.212*	0.923	0.53

11

Setting	Pipeline	Design	TJ+N	TJ+B	Gauss-LOO	Split	ANOVA-PI	Gauss-LGO	ASTM-PI	TJ+N, same batches	ICC
inst_15029	PDS	2×10	0.877*	unb.	0.761*	0.868*	0.890	0.994	0.298*	0.925	0.56
inst_15029	PDS	5×4	0.904	unb.	0.861*	0.896	0.924	0.952	0.335*	0.921	0.54
inst_15029	PDS	10×2	0.924	0.974	0.908	0.907	0.942	0.943	0.386*	0.928	0.55
inst_15029	PDS	15×1	0.959	0.959	0.917	unb.	0.926	0.917	0.368*	0.964	0.54
inst_15029	PDS	20×1	0.932	0.932	0.923	0.911	0.931	0.923	0.415*	0.933	0.53
inst_15029	SBC	chrono 5×4	0.735*	unb.	0.681*	0.767*	0.908	0.896	0.320*	0.913	0.36
inst_15029	SBC	2×10	0.723*	unb.	0.709*	0.742*	0.923	0.997	0.707*	0.908	0.61
inst_15029	SBC	5×4	0.827*	unb.	0.823*	0.839*	0.913	0.917	0.748*	0.915	0.61
inst_15029	SBC	10×2	0.888*	0.945	0.879*	0.897	0.919	0.908	0.771*	0.923	0.61
inst_15029	SBC	15×1	0.932	0.932	0.892	unb.	0.907	0.892	0.772*	0.956	0.60
inst_15029	SBC	20×1	0.908	0.908	0.899	0.900	0.911	0.899	0.781*	0.924	0.59
season	NONE	chrono 5×4	0.914	unb.	0.923	0.915	0.916	0.973	0.746*	0.906	0.64
season	NONE	2×10	0.870*	unb.	0.871*	0.878*	0.882*	1.000	0.750*	0.908	0.56
season	NONE	5×4	0.897	unb.	0.900	0.903	0.912	0.956	0.754*	0.910	0.55
season	NONE	10×2	0.900	0.950	0.898	0.901	0.904	0.918	0.747*	0.911	0.55
season	NONE	15×1	0.946	0.946	0.910	unb.	0.905	0.910	0.762*	0.945	0.47
season	SBC	chrono 5×4	0.853*	unb.	0.860*	0.863*	0.936	0.951	0.725*	0.907	0.63
season	SBC	2×10	0.676*	unb.	0.675*	0.693*	0.926	0.999	0.634*	0.913	0.58
season	SBC	5×4	0.824*	unb.	0.830*	0.838*	0.941	0.944	0.689*	0.920	0.56
season	SBC	10×2	0.884*	0.950	0.878*	0.879*	0.930	0.917	0.714*	0.930	0.55
season	SBC	15×1	0.933	0.933	0.898	unb.	0.920	0.898	0.721*	0.953	0.49

Table S12. Simpler interval formulas

Table S12. Simpler interval formulas on identical standards (200 repetitions; standards drawn at random).

Data set	Pipeline	Method	Too narrow α=0.05	Too narrow α=0.10	Too narrow α=0.20	Median IS relative to TJ+N (α=0.10)
corn	pds	ASTM-T	0/24	0/24	0/24	1.017
corn	pds	Boot-T	7/24	3/24	2/24	0.993
corn	pds	Gauss-LOO	6/24	0/24	0/24	0.941
corn	pds	Gauss-LOO-N	0/24	0/24	0/24	0.913
corn	pds	SCP-T-split-N	0/24	0/24	0/24	1.185
corn	pds	TJ+	0/24	0/24	0/24	1.021
corn	pds	TJ+N	0/24	0/24	0/24	1.0
corn	sbc	ASTM-T	0/24	0/24	0/24	1.524
corn	sbc	Boot-T	24/24	19/24	12/24	0.968
corn	sbc	Gauss-LOO	8/24	1/24	0/24	0.957
corn	sbc	Gauss-LOO-N	7/24	1/24	0/24	0.951
corn	sbc	OLS-PI	5/24	1/24	0/24	0.955

Data set	Pipeline	Method	Too narrow α=0.05	Too narrow α=0.10	Too narrow α=0.20	Median IS relative to TJ+N (α=0.10)
corn	sbc	SCP-T-split-N	0/24	0/24	0/24	1.133
corn	sbc	TJ+	0/24	0/24	0/24	1.013
corn	sbc	TJ+N	0/24	0/24	0/24	1.0
mango	sbc	ASTM-T	0/4	0/4	0/4	1.01
mango	sbc	Boot-T	4/4	4/4	3/4	1.006
mango	sbc	Gauss-LOO	0/4	0/4	1/4	1.01
mango	sbc	Gauss-LOO-N	0/4	0/4	2/4	0.995
mango	sbc	OLS-PI	0/4	0/4	3/4	1.007
mango	sbc	SCP-T-split-N	0/4	0/4	0/4	1.091
mango	sbc	TJ+	0/4	0/4	0/4	1.015
mango	sbc	TJ+N	0/4	0/4	0/4	1.0
mango_inst	pds	ASTM-T	0/12	0/12	0/12	1.126
mango_inst	pds	Boot-T	0/12	0/12	0/12	1.223
mango_inst	pds	Gauss-LOO	0/12	0/12	0/12	1.123
mango_inst	pds	Gauss-LOO-N	0/12	0/12	3/12	0.953
mango_inst	pds	SCP-T-split-N	0/12	0/12	0/12	1.222
mango_inst	pds	TJ+	0/12	0/12	0/12	1.197
mango_inst	pds	TJ+N	0/12	0/12	0/12	1.0
mango_inst	sbc	ASTM-T	4/12	0/12	0/12	0.966
mango_inst	sbc	Boot-T	12/12	11/12	6/12	0.964
mango_inst	sbc	Gauss-LOO	4/12	0/12	0/12	0.966
mango_inst	sbc	Gauss-LOO-N	6/12	0/12	0/12	0.986
mango_inst	sbc	OLS-PI	3/12	0/12	0/12	0.964
mango_inst	sbc	SCP-T-split-N	0/12	0/12	0/12	1.08
mango_inst	sbc	TJ+	0/12	0/12	0/12	0.982
mango_inst	sbc	TJ+N	0/12	0/12	0/12	1.0
tablets	pds	ASTM-T	4/12	0/12	1/12	1.052
tablets	pds	Boot-T	8/12	7/12	8/12	1.031
tablets	pds	Gauss-LOO	5/12	0/12	1/12	1.058
tablets	pds	Gauss-LOO-N	2/12	1/12	0/12	0.992
tablets	pds	SCP-T-split-N	0/12	0/12	0/12	1.067
tablets	pds	TJ+	0/12	0/12	0/12	1.057
tablets	pds	TJ+N	0/12	0/12	0/12	1.0
tablets	sbc	ASTM-T	8/12	1/12	1/12	1.005
tablets	sbc	Boot-T	10/12	8/12	6/12	0.991
tablets	sbc	Gauss-LOO	8/12	1/12	1/12	1.005
tablets	sbc	Gauss-LOO-N	8/12	1/12	1/12	0.988
tablets	sbc	OLS-PI	8/12	1/12	1/12	0.995
tablets	sbc	SCP-T-split-N	0/12	0/12	0/12	1.072
tablets	sbc	TJ+	0/12	0/12	0/12	1.017
tablets	sbc	TJ+N	0/12	0/12	0/12	1.0

13

Table S13. Closeness to nominal and per-set reliability

Table S13. Main scenarios with random standards (corn, tablets, mango season shift and mango instruments; 64 scenario-pipeline combinations). Non-inferior: lower limit of the 95% CI of coverage ≥ nominal − 0.02. Last column: share of standard sets (repetitions) with realised coverage ≥ 0.85 on the test samples, median (minimum) over scenarios. Unbounded intervals count as covering.

Method	Non-inferior α=0.05	α=0.10	α=0.20	Training-conditional share, median (min), α=0.10
TJ+N	64/64	64/64	64/64	0.88 (0.74)
TJ+	64/64	64/64	64/64	0.88 (0.76)
Gauss-LOO	62/64	63/64	62/64	0.82 (0.73)
SCP-T-split-N	64/64	64/64	64/64	0.91 (0.74)
ASTM-PI	11/64	11/64	11/64	0.00 (0.00)

Table S14. Harvest-maturity case study, all settings

Table S14. Screening at 14% dry matter with 90% intervals, fruit from consignments without standards (200 repetitions). Each cell: percentage of immature fruit passed as mature / percentage of all fruit referred for re-measurement. Bootstrap CIs: results/summary/E14_summary.csv.

Setting	Pipeline	Design	Immature %	Point	ASTM-PI	Gauss-LOO	TJ+N	TJ+B
inst_15014	PDS	2×10	25	39.8 / 0	25.6 / 24	8.0 / 63	2.3 / 78	0.0 / 100
inst_15014	PDS	10×2	25	36.7 / 0	20.4 / 27	2.3 / 78	1.4 / 79	0.4 / 89
inst_15014	PDS	20×1	24	31.9 / 0	16.1 / 28	1.6 / 78	1.3 / 73	1.3 / 73
inst_15014	SBC	2×10	25	31.5 / 0	8.2 / 34	7.8 / 36	7.8 / 37	0.0 / 100
inst_15014	SBC	10×2	25	31.2 / 0	5.4 / 38	2.1 / 51	2.1 / 52	0.7 / 61
inst_15014	SBC	20×1	24	27.6 / 0	5.0 / 38	1.8 / 52	1.9 / 53	1.9 / 53
inst_15016	PDS	2×10	24	49.2 / 0	36.2 / 23	12.0 / 69	5.3 / 80	0.0 / 100
inst_15016	PDS	10×2	24	58.7 / 0	44.4 / 24	5.0 / 88	4.8 / 86	1.1 / 94
inst_15016	PDS	20×1	24	65.4 / 0	51.3 / 24	5.2 / 91	4.3 / 88	4.3 / 88
inst_15016	SBC	2×10	24	44.5 / 0	12.0 / 40	9.8 / 44	9.4 / 49	0.0 / 100
inst_15016	SBC	10×2	24	38.1 / 0	8.4 / 41	3.8 / 54	4.6 / 57	2.0 / 69
inst_15016	SBC	20×1	24	37.6 / 0	8.4 / 41	3.7 / 53	4.5 / 57	4.5 / 57
inst_15029	PDS	2×10	24	69.3 / 0	52.8 / 22	19.9 / 67	12.0 / 77	0.0 / 100
inst_15029	PDS	10×2	24	75.0 / 0	53.8 / 25	6.0 / 81	7.9 / 79	2.3 / 90
inst_15029	PDS	20×1	23	74.2 / 0	49.5 / 26	3.0 / 82	6.8 / 78	6.8 / 78
inst_15029	SBC	2×10	24	44.0 / 0	12.4 / 37	12.8 / 41	11.5 / 43	0.0 / 100
inst_15029	SBC	10×2	24	41.5 / 0	9.9 / 39	5.8 / 50	5.0 / 52	2.4 / 62
inst_15029	SBC	20×1	23	39.7 / 0	9.9 / 38	4.6 / 52	4.1 / 54	4.1 / 54

Setting	Pipeline	Design	Immature %	Point	ASTM-PI	Gauss-LOO	TJ+N	TJ+B
season	SBC	2×10	14	42.1 / 0	10.9 / 23	9.8 / 25	10.6 / 25	0.0 / 100
season	SBC	10×2	14	34.3 / 0	4.3 / 26	0.9 / 34	1.1 / 34	0.3 / 41

Table S15. Confirmatory study, every setting

Table S15. Confirmatory study on 14 handheld-unit × season settings (slope/bias correction; α = 0.10; 300 repetitions). Coverage for fruit from consignments without standards; * flagged as too narrow; unb.: unbounded. Planning model: predicted coverage of sample-level TJ+ from the design and the observed ICC.

Unit / season	Design	TJ+N	Gauss-LOO	OLS-PI	ASTM-PI	TJ+B	Gauss-LGO	ANOVA-PI	Planning model (TJ+)	ICC	RMSEP
14002 / 2016	2×10	0.781*	0.763*	0.774*	0.663*	unb.	1.000	0.926	0.833	0.38	1.40
14002 / 2016	5×4	0.878*	0.875*	0.880*	0.721*	unb.	0.957	0.948	0.881	0.37	1.27
14002 / 2016	10×2	0.888*	0.895	0.898	0.728*	0.946	0.924	0.933	0.897	0.36	1.25
14002 / 2017	2×10	0.697*	0.684*	0.700*	0.698*	unb.	0.999	0.938	0.761	0.60	1.36
14002 / 2017	5×4	0.823*	0.812*	0.815*	0.749*	unb.	0.923	0.920	0.857	0.61	1.22
14002 / 2017	10×2	0.864*	0.858*	0.858*	0.762*	0.939	0.894	0.908	0.890	0.59	1.17
15015 / 2016	2×10	0.810*	0.809*	0.817*	0.913	unb.	0.999	0.928	0.836	0.36	0.90
15015 / 2016	5×4	0.865*	0.865*	0.867*	0.933	unb.	0.936	0.920	0.883	0.35	0.85
15015 / 2016	10×2	0.890*	0.886*	0.886*	0.942	0.950	0.909	0.916	0.898	0.34	0.82
15023 / 2016	2×10	0.758*	0.760*	0.768*	0.827*	unb.	1.000	0.945	0.831	0.38	0.98
15023 / 2016	5×4	0.855*	0.853*	0.855*	0.864*	unb.	0.946	0.935	0.882	0.37	0.92
15023 / 2016	10×2	0.899	0.898	0.897	0.881*	0.958	0.928	0.937	0.898	0.35	0.88
15023 / 2017	2×10	0.755*	0.738*	0.755*	0.821*	unb.	0.999	0.918	0.816	0.44	0.99
15023 / 2017	5×4	0.866*	0.854*	0.858*	0.866*	unb.	0.945	0.936	0.877	0.42	0.89
15023 / 2017	10×2	0.896	0.888*	0.889*	0.888*	0.952	0.916	0.925	0.896	0.40	0.85
15038 / 2016	2×10	0.700*	0.691*	0.695*	0.696*	unb.	0.999	0.928	0.776	0.56	1.39
15038 / 2016	5×4	0.824*	0.821*	0.822*	0.743*	unb.	0.925	0.922	0.862	0.57	1.28
15038 / 2016	10×2	0.888*	0.882*	0.882*	0.759*	0.946	0.915	0.927	0.892	0.55	1.21
16033 / 2017	2×10	0.745*	0.734*	0.748*	0.856*	unb.	1.000	0.932	0.798	0.50	0.95
16033 / 2017	5×4	0.843*	0.830*	0.836*	0.895	unb.	0.932	0.920	0.871	0.48	0.86
16033 / 2017	10×2	0.892	0.892	0.891	0.920	0.957	0.925	0.934	0.895	0.45	0.80
16041 / 2017	2×10	0.724*	0.718*	0.736*	0.733*	unb.	0.998	0.938	0.793	0.51	1.28
16041 / 2017	5×4	0.853*	0.847*	0.851*	0.799*	unb.	0.944	0.936	0.873	0.47	1.12
16041	10×2	0.890*	0.886*	0.885*	0.821*	0.951	0.915	0.924	0.895	0.45	1.07

15

Unit / season	Design	TJ+N	Gauss-LOO	OLS-PI	ASTM-PI	TJ+B	Gauss-LGO	ANOVA-PI	Planning model (TJ+)	ICC	RMSEP
/ 2017											
16041	20×1	0.916	0.912	0.912	0.834*	0.916	0.912	0.924	0.905	0.37	1.01
/ 2017											
16048	2×10	0.788*	0.760*	0.764*	0.693*	unb.	0.997	0.915	0.819	0.43	1.33
/ 2018											
16048	5×4	0.873*	0.856*	0.859*	0.737*	unb.	0.939	0.931	0.879	0.40	1.22
/ 2018											
16048	10×2	0.886	0.867*	0.867*	0.747*	0.939	0.899	0.907	0.897	0.37	1.18
/ 2018											
17033	2×10	0.700*	0.679*	0.695*	0.634*	unb.	0.999	0.915	0.789	0.53	1.61
/ 2017											
17033	5×4	0.845*	0.831*	0.833*	0.707*	unb.	0.933	0.928	0.867	0.52	1.40
/ 2017											
17033	10×2	0.896	0.882*	0.882*	0.731*	0.953	0.914	0.924	0.893	0.51	1.33
/ 2017											
17033	20×1	0.908	0.907	0.906	0.740*	0.908	0.907	0.921	0.905	0.48	1.30
/ 2017											
17052	2×10	0.672*	0.659*	0.677*	0.759*	unb.	0.997	0.920	0.743	0.64	1.21
/ 2017											
17052	5×4	0.824*	0.819*	0.824*	0.820*	unb.	0.929	0.927	0.852	0.64	1.05
/ 2017											
17052	10×2	0.877*	0.874*	0.872*	0.843*	0.947	0.913	0.924	0.889	0.64	0.99
/ 2017											
17057	2×10	0.790*	0.777*	0.781*	0.721*	unb.	0.996	0.901	0.836	0.36	1.28
/ 2018											
17057	5×4	0.864*	0.862*	0.859*	0.765*	unb.	0.926	0.917	0.882	0.37	1.17
/ 2018											
17057	10×2	0.904	0.905	0.899	0.782*	0.955	0.922	0.928	0.898	0.34	1.13
/ 2018											
30004	2×10	0.788*	0.782*	0.799*	0.888*	unb.	0.998	0.929	0.839	0.35	0.98
/ 2016											
30004	5×4	0.861*	0.860*	0.863*	0.918	unb.	0.943	0.929	0.886	0.31	0.91
/ 2016											
30004	10×2	0.887*	0.891	0.890*	0.931	0.952	0.917	0.923	0.899	0.29	0.86
/ 2016											
30004	2×10	0.645*	0.630*	0.641*	0.726*	unb.	0.999	0.916	0.738	0.66	1.32
/ 2017											
30004	5×4	0.810*	0.808*	0.814*	0.780*	unb.	0.931	0.933	0.848	0.67	1.19
/ 2017											
30004	10×2	0.868*	0.871*	0.874*	0.807*	0.940	0.912	0.924	0.888	0.66	1.12
/ 2017											

Table S16. Guard bands and interval rules

Table S16. Harvest-maturity screening at 14% dry matter (slope/bias correction; mean over the season shift and three handheld-instrument pairs; 200 repetitions). Guard-k: pass if the prediction exceeds 14% by k × leave-one-out RMSE, fail if it is more than that below, refer otherwise. TJ+N@a, Gauss-LOO@a: intervals at miscoverage a.

Design	Rule	Immature passed, %	Referred, %
10×2	Point	36.3	0.0
10×2	Guard-0.25	27.4	8.2
10×2	Guard-0.5	20.0	16.2
10×2	Guard-0.75	14.2	23.7
10×2	Guard-1	10.0	30.6
10×2	Gauss-LOO@0.3	9.2	32.3
10×2	TJ+N@0.3	9.1	33.7
10×2	ASTM-PI	7.0	36.0

Design	Rule	Immature passed, %	Referred, %
10×2	Guard-1.25	7.0	36.9
10×2	Gauss-LOO@0.2	6.3	38.7
10×2	TJ+N@0.2	6.1	40.2
10×2	Guard-1.5	4.8	42.6
10×2	Gauss-LOO	3.2	47.3
10×2	Guard-1.75	3.1	47.7
10×2	TJ+N	3.2	48.7
10×2	Guard-2	1.9	52.4
10×2	Gauss-LOO@0.05	1.6	54.1
10×2	TJ+N@0.05	1.6	56.9
10×2	TJ+H-N	1.4	58.3
10×2	Guard-2.5	0.7	61.0
10×2	Guard-3	0.3	68.6
20×1	Point	35.0	0.0
20×1	Guard-0.25	26.7	8.6
20×1	Guard-0.5	19.8	17.4
20×1	Guard-0.75	14.3	25.8
20×1	Guard-1	10.3	33.5
20×1	Gauss-LOO@0.3	9.5	35.4
20×1	TJ+N@0.3	9.5	36.8
20×1	ASTM-PI	7.8	38.9
20×1	Guard-1.25	7.5	40.6
20×1	Gauss-LOO@0.2	6.7	42.6
20×1	TJ+N@0.2	6.7	44.3
20×1	Guard-1.5	5.2	46.9
20×1	Gauss-LOO	3.4	52.3
20×1	Guard-1.75	3.3	52.8
20×1	TJ+H-N	3.5	54.5
20×1	TJ+N	3.5	54.5
20×1	Guard-2	1.9	58.0
20×1	Gauss-LOO@0.05	1.5	59.9
20×1	TJ+N@0.05	1.5	63.4
20×1	Guard-2.5	0.6	67.4
20×1	Guard-3	0.1	75.3
2×10	Point	40.5	0.0
2×10	Guard-0.25	34.9	5.6
2×10	Guard-0.5	29.3	11.4
2×10	Guard-0.75	24.4	16.9
2×10	Guard-1	19.7	22.3
2×10	Gauss-LOO@0.3	18.7	23.7
2×10	TJ+N@0.3	18.4	24.6
2×10	Guard-1.25	15.9	27.5
2×10	Gauss-LOO@0.2	14.8	29.1
2×10	TJ+N@0.2	14.6	29.9
2×10	Guard-1.5	12.6	32.4
2×10	ASTM-PI	10.9	33.7
2×10	Gauss-LOO	10.1	36.7
2×10	Guard-1.75	9.9	37.0
2×10	TJ+N	9.8	38.5
2×10	Guard-2	7.7	41.4
2×10	Gauss-LOO@0.05	7.0	43.0
2×10	TJ+N@0.05	6.3	46.9
2×10	Guard-2.5	4.7	49.4
2×10	Guard-3	2.9	56.3
2×10	TJ+H-N	0.0	100.0

Fig. S1

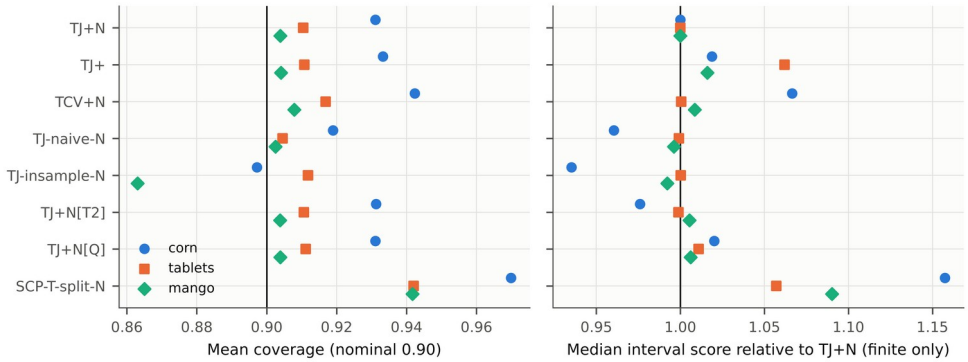


Fig. S1. Ablation of TJ+N: mean coverage (left) and median interval score relative to TJ+N (right).

Fig. S2

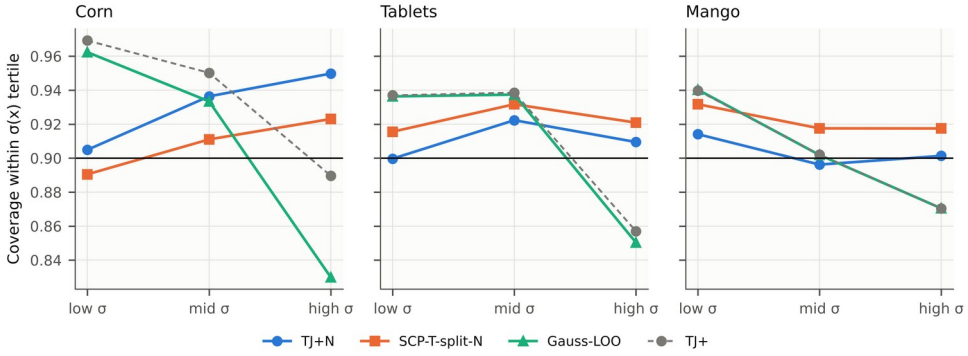


Fig. S2. Coverage within thirds of the difficulty σ(x) of the test spectra (main pipelines, α = 0.10).

Fig. S3

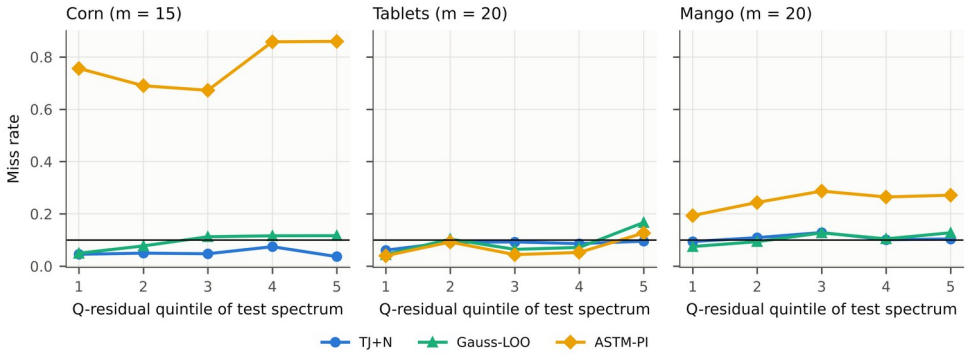


Fig. S3. Miss rate by Q-residual quintile of the test spectra (first 100 repetitions).

18

Fig. S4

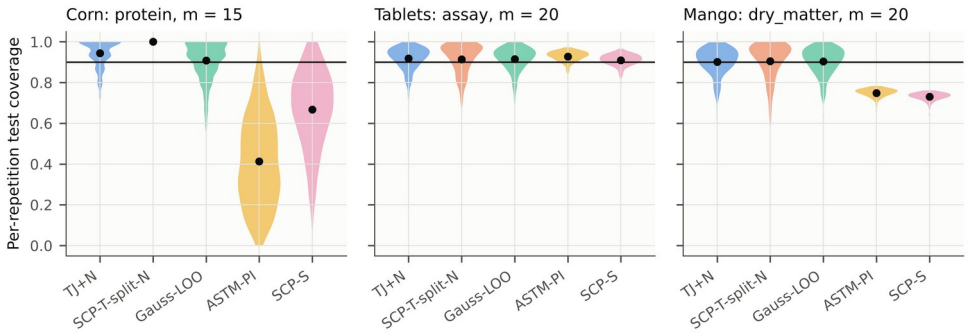


Fig. S4. Distribution over repetitions of the realised coverage for one scenario per data set; the guarantee concerns the mean.

Fig. S5

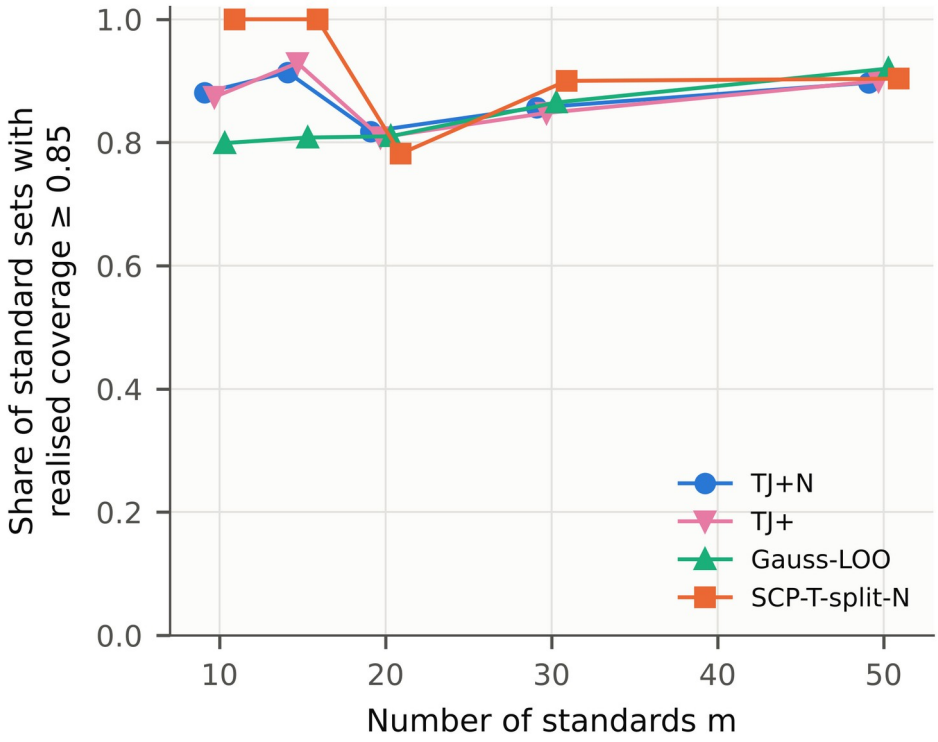


Fig. S5. Share of standard sets whose realised coverage reached 0.85, by number of standards (median over scenarios).

19

Fig. S6

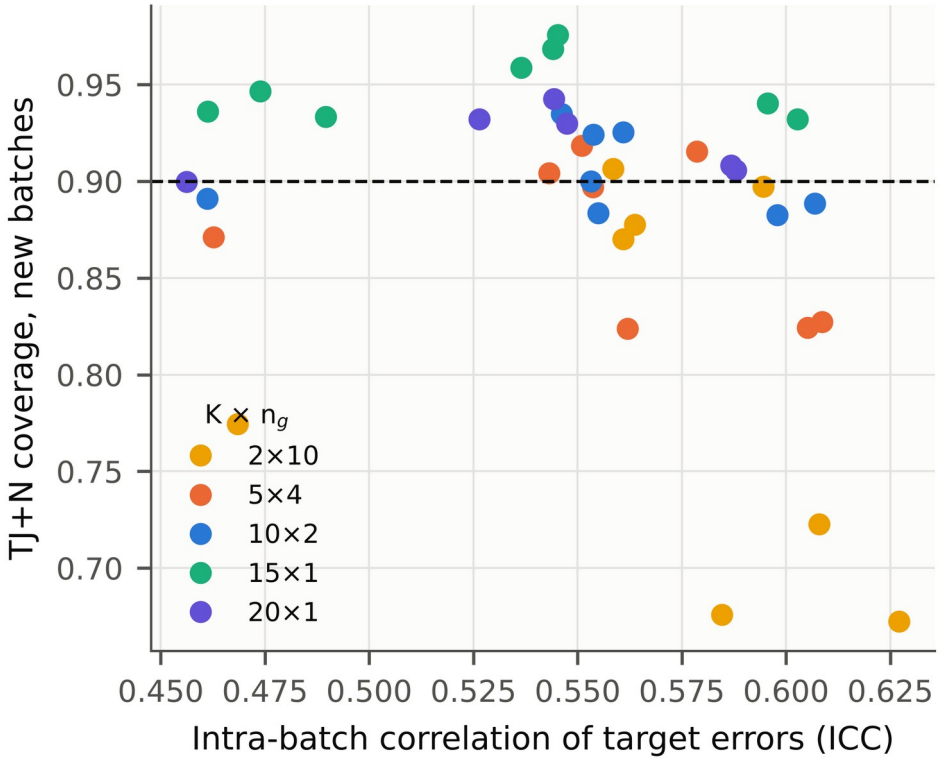


Fig. S6. TJ+N coverage for new consignments against the intra-consignment correlation of the errors, by design. Within the narrow ICC range observed, the design (number of consignments) rather than the ICC determined coverage.

Fig. S7

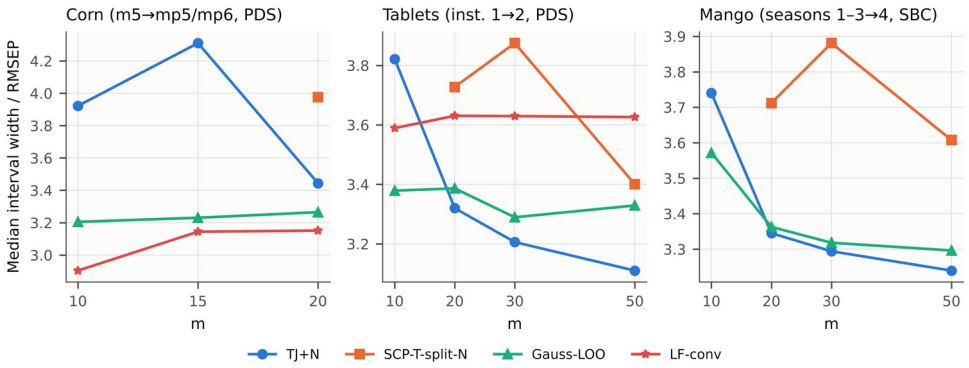


Fig. S7. Median interval width relative to RMSEP for methods that were not too narrow.

20

Fig. S8

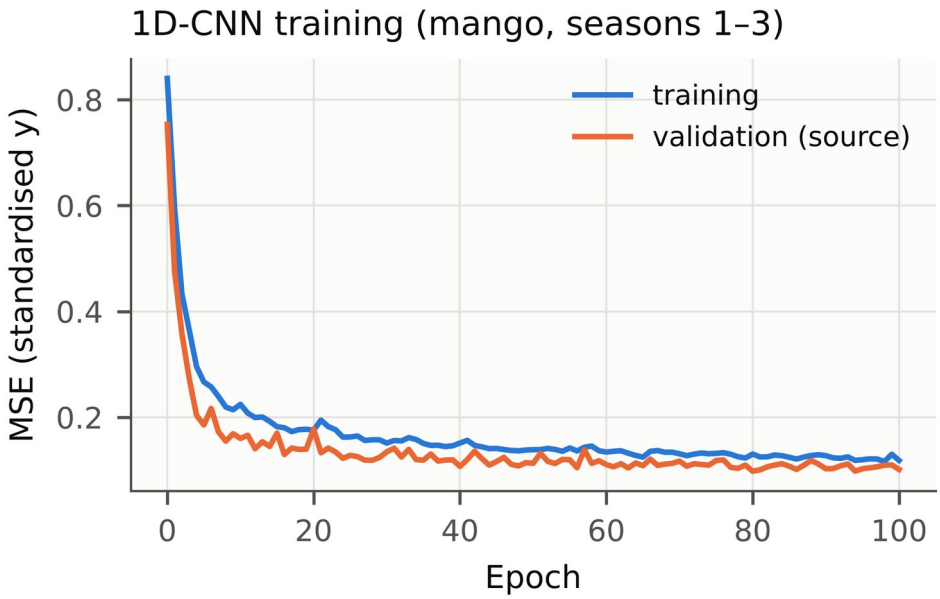


Fig. S8. Training and validation loss of the 1D-CNN (mango, seasons 1–3).

Fig. S9

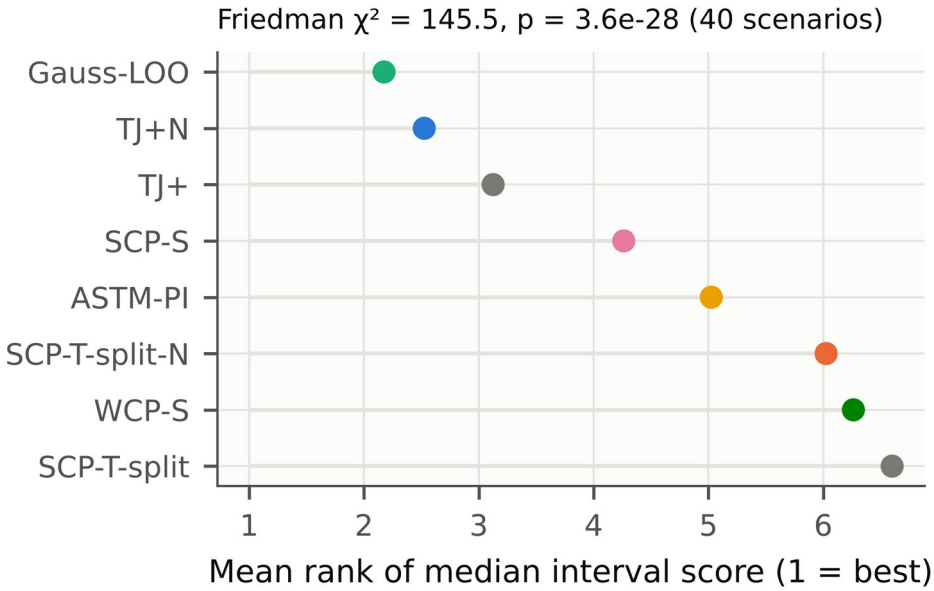


Fig. S9. Mean Friedman ranks of the scenario median interval score over the 40 main scenarios (descriptive).

21

Fig. S10

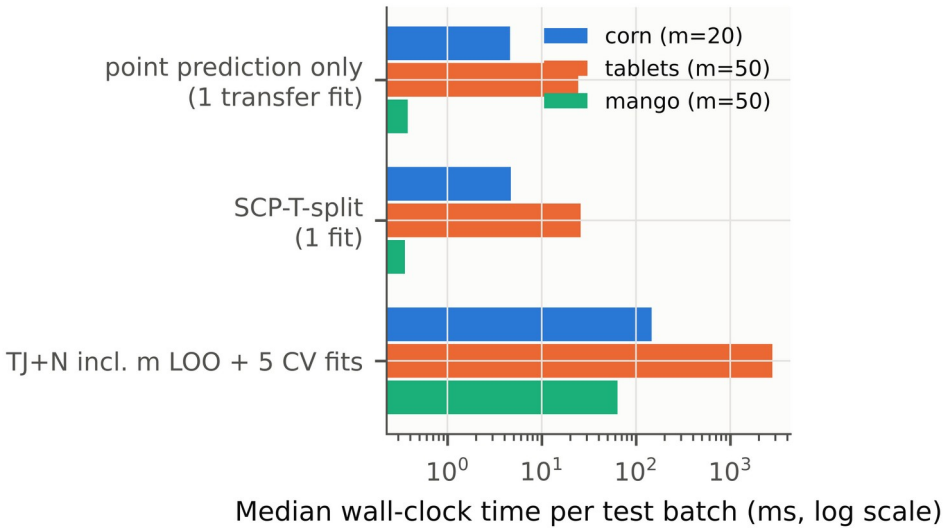


Fig. S10. Wall-clock time per test batch (one CPU core, median of 20 runs).

22